4.3 Mendelian Genetics and Inheritance Patterns

Unit 4: Genetic Processes

Mendel's Work

Gregor Mendel (1860s) studied inheritance in pea plants by controlling crosses and counting offspring. He proposed particulate factors (now called genes/alleles) that are passed from parents to offspring, not blended.

Key Terminology

Term	Definition
Gene	A segment of DNA that codes for a specific trait. Located at a locus on a chromosome.
Allele	Alternative forms of a gene (e.g., B for brown eyes, b for blue eyes).
Dominant	The allele that is expressed when paired with a different allele. Written with a capital letter.
Recessive	The allele expressed only when present in two copies (homozygous recessive). Written with lowercase.
Homozygous	Two identical alleles (BB or bb).
Heterozygous	Two different alleles (Bb). Also called a carrier for recessive traits.
Genotype	The allele combination of an organism (e.g., Bb).
Phenotype	The observable trait expressed (e.g., brown eyes).

Mendel's Laws

Term	Definition
Law of Segregation	Each organism has two alleles for each trait; during gamete formation, these separate so each gamete carries only one allele.
Law of Independent Assortment	Alleles of different genes segregate independently during gamete formation (true only for genes on different chromosomes).

Punnett Squares and Ratios

■ Tip

Testcross: cross an unknown genotype with a homozygous recessive (bb). If any offspring show the recessive phenotype, the unknown parent must be heterozygous.

Monohybrid cross (one trait) between two heterozygotes (Bb × Bb):

BB : Bb : bb = 1 : 2 : 1 (genotypic ratio)

Brown : Blue = 3 : 1 (phenotypic ratio)

Dihybrid cross (two traits): BbRr × BbRr → 9:3:3:1 phenotypic ratio (9 dominant-dominant : 3 dominant-recessive : 3 recessive-dominant : 1 recessive-recessive)

Extensions of Mendelian Genetics

Term	Definition
Incomplete dominance	Heterozygote has intermediate phenotype. Example: red × white carnations → pink (RW). Ratio 1:2:1.
Codominance	Both alleles expressed equally in heterozygote. Example: blood type AB. Ratio 1:2:1.
Multiple alleles	More than two alleles exist for one gene in the population. Example: ABO blood type (I^A, I^B, i).
Polygenic inheritance	Multiple genes contribute to one trait. Results in a range of phenotypes (continuous variation). Example: skin colour, height.
Epistasis	One gene masks the expression of another gene. Example: coat colour in Labrador retrievers.